The Next Quantum Game

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Abstract

This report presents the design, development and analysis of our educational game aimed at introducing key concepts of quantum computing to a broad audience beyond the academic community. The project seeks to make abstract quantum phenomena intuitive and engaging through interactive gameplay. We describe the motivation for creating the game, along with the key design and implementation choices made to balance conceptual accuracy with playability. These design decisions are evaluated against a defined set of success criteria using results from a player survey, which serves as the primary means of assessing the game's effectiveness. The analysis made on our survey provides quantitative and qualitative insights into player engagement, conceptual understanding, and overall learning experience. Finally, we discuss how these findings inform potential improvements and future directions for the game's development.

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1 Introduction

The field of quantum computing becomes increasingly relevant as quantum computing increases its usability[SW23] and receives increased funding[Gib19]. This is knowledge that academics and students in the field can catch up with through conventional means, like textbooks and lectures. However, since concepts in quantum computing often conflict with the classical intuition of individuals unfamiliar with the field, such audiences may find it difficult to grasp these ideas, particularly due to their highly mathematical nature[Jar18]. Therefore, we try to introduce these concepts through video game mechanics in order to make it intuitive for the general audience, as video games are engaging in nature and make people learn the mechanics in order to play through the video game[Sch16].

With this in mind, our goal is not just to have a fun game, but also to design a learning experience that conveys core principles of quantum computing. We have four criteria with which we measure our success, as we believe these criteria are needed for an educational game like ours to be called successful:

- Gameplay Engagement
- Conceptual Intuitiveness
- Scientific Accuracy
- Technical Implementation

The structure of this report is as follows: it begins with the Problem Description, outlining the context and elaborating on the success criteria. This is followed by the Development Overview, discussing key development choices and their outcomes. The Game Design & Implementation section then describes the game's core features and mechanics, after which we highlight the Interdisciplinary Aspects involved in our project. We then discuss our Teamwork. The Survey Analysis section presents the survey's design and a detailed discussion of the results. Finally, the report concludes with Assessment and Possible Improvements, and an overall Conclusion, summarising key insights and future directions.

2 Problem Description

2.1 Problem Context

The field of quantum computing has terms like qubits, superposition and entanglement for which we believe many readers of this paper are familiar with. However, many people take the sources of their explanations from places other than verified sources, as said sources offer simpler explanations, leading to them sometimes learning misinformation instead, as an explanation they understand is one they will remember[Seh24]. We therefore aim to offer explanations through gamification in order to offer people a truthful alternative[Sch16]. To ensure that, we do have to make the explanations in-game truthful or at the very least explain what can happen in real life and what cannot. Therefore, we aim to have an experience accessible for everyone through all academic skills, but not oversimplified, as people getting information from oversimplified narratives is a problem we aim to solve with this game as well. Therefore, our problem statement is the following:

Design and evaluate a new quantum game that uses core quantum computing concepts as gameplay mechanics to make quantum phenomena fun and intuitive to a wide audience.

Within the scope of our project, we aim to explain some concepts in the field of quantum computing. We want to make the game accessible to high school students and older, who are expected to understand the underlying concepts, while younger players may still find the game fun despite not fully relating it to real world concepts. Teaching the full mathematical formalism or building a commercial-grade game is out of the scope of this project.

Our work is part of the ongoing effort to make quantum computing more approachable. Some existing ways to achieve this include

- **Interactive Tools:** Like the works of Paul Falstad[Fal] and Physlet simulations[CBE+15] among others.
- **Interactive Explanations:** Like "Quantum Country"[MN19] and "Building your own Quantum Fourier Transform"[Gid14] among others.
- **Other Quantum Games:** Like Quantum Tic Tac Toe[Nie] and Quantum Odyssey[Jan22] among others.

An attempt to classify Quantum Games has been made in the work by Seskir et al[ZCS22].

2.2 Success Criteria

For our success criteria, we took inspiration from the criteria used by Quantum Games Hackathon by the Quantum AI Foundation[ZCS22]. This results in the following criteria:

- **Gameplay Engagement**: This is related to how much players enjoyed game. We score this on a scale of 1 to 5, with 1 meaning **Bored**, 2 meaning **A bit bored**, 3 meaning **Neutral**, 4 meaning **Enjoyed it**, and 5 meaning **Loved it**.
- **Conceptual Intuitiveness**: This is related to how much players learn. We score this on a scale of 1 to 5 with 1 meaning **Nothing**, 2 meaning **A little**, 3 meaning **A moderate amount**, 4 meaning **A lot**, and 5 meaning **A great deal**.
- **Scientific Accuracy**: This is related to how accurately the phenomena were represented. We score this on a scale of 1 to 5, with 1 meaning **Completely inaccurate**, 2 meaning **Misses some key aspects**, 3 meaning **Room for misinterpretations**, 4 meaning **Misses minor aspects**, and 5 meaning **Extremely accurate**.
- **Technical Implementation**: This is related to the game quality. We score this on a scale of 1 to 5, with 1 meaning **Very poor**, 2 meaning **Poor**, 3 meaning **Average**, 4 meaning **Good**, and 5 meaning **Excellent**.

Compared to the paper[ZCS22] however, we have made some minor changes. First of all, gameplay engagement is treated separately from its educational ability, as a game can be engaging without having an educational purpose and vice versa. Secondly, originality has been removed and replaced by conceptual intuitiveness, as that is about educational potential. One of the reasons we have changed our success criteria is that our game will not be judged by qualified judges. Instead, it will be judged by players playing our game via a survey.

3 Development Overview

This section elaborates on the key decisions made during the development of our quantum game, along with their motivations and outcomes.

3.1 Development Choices

Game Type (Platformer): Our first major decision was regarding the type of game to develop. After brainstorming a few concepts, including a quantum-based variant of *Angry Birds*, we ultimately chose to design a platformer game. This genre provided an intuitive way to represent quantum concepts such as two-level systems (here, qubits), superposition, and measurement within a familiar gameplay framework. Also, it allowed us to introduce more quantum-related phenomena, such as circuit-based puzzles and two-qubit systems (and entanglement), across different levels of the game.

Version Control (GitHub): Given the collaborative nature of our project, our next step was to create a version control system for our code. We decided to make a GitHub repository, allowing us to safely push individual contributions and resolve code conflicts efficiently. During the initial phase of development, we would push directly into the `main` branch since the changes made were very foundational. Eventually, as the project matured, we shifted towards creating feature branches and requiring approvals on merge requests for code commits, which helped ensure code quality, and support collaborative development at scale.

Game Engine (Godot): Finally, we selected a suitable game engine. Among popular options like Unity, Unreal Engine, and Godot, we chose Godot due to its open-source nature, availability of helpful platformer tutorials (especially the one by Brackeys[Bra]), and ease of scripting in GDScript (a language syntactically similar to Python), which facilitated quick development and team collaboration. Godot also has a small install size (100-200 MB) and extremely fast startup and iteration times, while Unity and Unreal are much heavier (multiple GBs) and slower to compile or launch. Godot's node-based architecture also aligned well with our modular design goals for implementing quantum features, for example, circuit puzzles. The node for circuit puzzles has many children, but we were able to import the same node to different levels of the game and customise it locally when required.

3.2 Outcomes

These workflow decisions resulted in a game framework that is both conceptually grounded in quantum computing and technically robust. The platformer genre provided a familiar yet flexible medium for representing quantum principles through gameplay mechanics. The Git-based workflow fostered an efficient and organised collaborative process, minimising merge conflicts and improving overall code quality. Finally, using Godot allowed us to maintain a clean, modular codebase and iterate rapidly, enabling the seamless integration of new quantum-inspired mechanics and visual elements as the project evolved.

4 Game Design & Implementation

This section describes the implementation details for our quantum game. It highlights the quantum phenomena chosen to implement in the game, and the methods used to represent them visually and interactively. The initial portion of the game is organised into Levels 0, 1, and 2; each progressively introduces new gameplay mechanics while reinforcing those presented in earlier stages. Finally, the game features a dedicated "Challenge Level" designed to provide players with a more complex and engaging test of their understanding and skills.

4.1 Gameplay Mechanics and Scoring

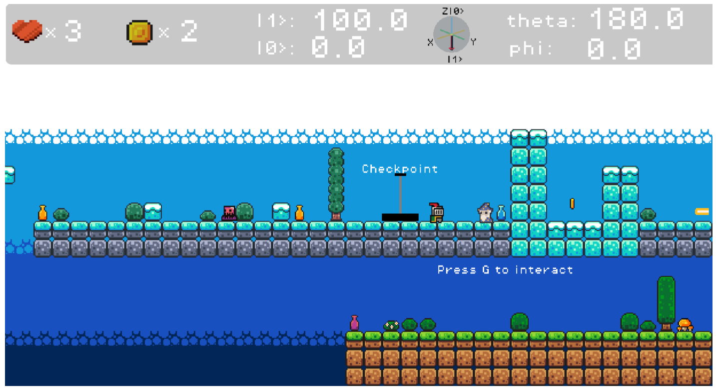


Figure 1: Player near a *Checkpoint*. On the top left, the number of hearts and coins currently held by the player are displayed.

Our game includes a system of *Checkpoints*. When the player loses a life (for instance, by colliding with an enemy or falling off the platform), they respawn at the most recently activated checkpoint, allowing them to continue from a nearby position rather than restarting the entire level. However, if the player's health (represented by hearts) is completely depleted, the level must be restarted from the beginning. Players can also regain health: collecting three coins restores one heart to their life.

During playtesting, we recognised the importance of balancing the frequency of checkpoints. Too many checkpoints made the gameplay feel trivial and reduced the sense of risk, while too few made the experience frustrating and repetitive. We therefore adopted a middle ground, ensuring that each checkpoint felt like a meaningful reward for progress rather than an automatic save. This design choice aligns with general principles of game balancing, which aim to maintain a sense of tension, challenge, and achievement throughout the player's experience [BG20]. Similarly, the "three coins for one heart" mechanic was introduced to further reinforce this balance, rewarding exploration and collection.

Figure 2: Screen after a level is cleared. The final score is displayed here.

Upon completing a level, a score is calculated for the user based on the time taken to complete the level, the number of hearts remaining, and the number of coins collected. Each user's own best score is saved, allowing them to compare and improve upon their existing scores.

stuff
more stuff

The score for each level is computed according to the following formula:

$$S_{\text{final}} = (10 \times S_{\text{coins}} + 50 \times H_{\text{remaining}}) \times M_{\text{time}}$$

where:

- S_{coins} is the number of coins collected in the level.
- $H_{\text{remaining}}$ is the number of hearts remaining upon completion.
- M_{time} represents the *multiplier*, which rewards faster completion while preventing extreme scoring imbalances.

The *multiplier* is defined as:

$$M_{\text{time}} = \text{clamp}\left(\frac{L_{\text{limit}}}{T_{\text{elapsed}}},\ 0.5,\ 5\right)$$

where,

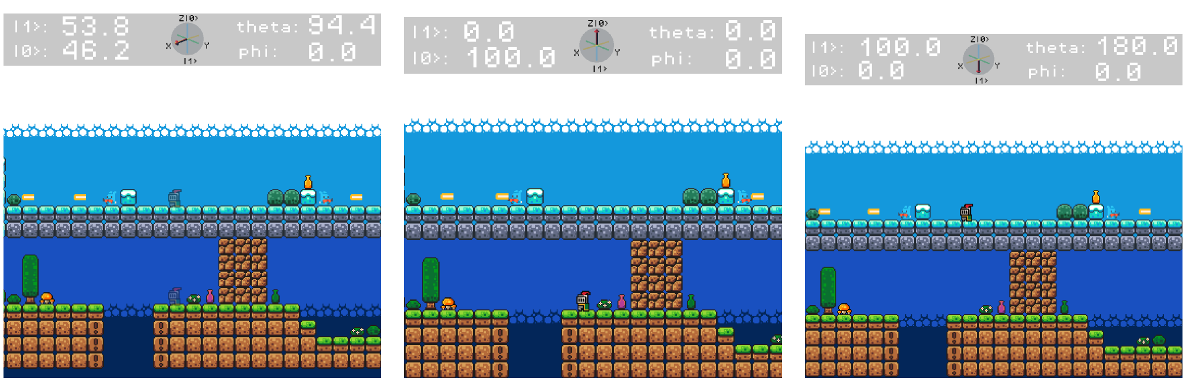
- T_{elapsed} is the total time taken (in seconds) to complete the level.
- L_{limit} is a level-specific reference time threshold:

$$L_{\text{limit}} = \begin{cases} 450, & \text{for Level 0} \\ 900, & \text{for Level 1} \\ 1200, & \text{for Level 2} \\ 7000, & \text{for the Challenge Level.} \end{cases}$$

Here, clamp(x, a, b) ensures that the value of x remains within the interval $[a, b]$, i.e., values below a are set to a, and values above b are set to b.

In essence, the player's score increases with the number of coins collected and hearts preserved, while also rewarding faster completion times. The clamping factor prevents extremely short or long completion times from disproportionately affecting the final score. This formulation was chosen to maintain a sense of challenge and reward, which aligns with the broader principles of game balancing as mentioned before, where fairness and player progression are prioritised for enhancing the overall experience of the game [BG20].

4.2 Core Gameplay



(a) Player enters a superposition. (b) Player measures to state $|0\rangle$. (c) Player measures to state $|1\rangle$.

Figure 3: Key quantum interactions in the game: entering superposition and performing a measurement.

The core gameplay is built on the mechanics of a traditional platformer game. The player moves horizontally using the arrow keys (or alternatively, the `A` and `D` keys), while the spacebar (or the `W` key) enables jumping. The player is also subject to constant gravity and has a configurable movement speed to balance responsiveness. However, unlike most platformer games, enemies cannot be defeated, only avoided.

The gameplay environment consists of two layers, representing the two basis states of a qubit: the bottom layer corresponds to $|0\rangle$ and the top layer to $|1\rangle$. The user can manipulate the player's quantum state through the single-qubit rotation gates $\mathrm{R_X}$, $\mathrm{R_Y}$, and $\mathrm{R_Z}$ by pressing the `X`, `Y`, and `Z` keys, respectively. Performing these rotations allows the player to generate any single-qubit state on the Bloch sphere, since $\mathrm{R_X}$, $\mathrm{R_Y}$, and $\mathrm{R_Z}$ form a universal set for single-qubit operations [NC10].

Applying these rotations serves a gameplay purpose: the player may need to change their superposition to increase their probability of appearing in one layer or the other. For example, to avoid an obstacle in $|0\rangle$ or collect a coin in $|1\rangle$. Superposition between $|0\rangle$ and $|1\rangle$ is visually represented using opacity: higher opacity in one layer indicates a higher probability of collapsing to that state.

To help players visualise their quantum state in real time, the Head-Up Display (HUD) (as can be seen in Fig 3) provides:

- The instantaneous probabilities of collapsing to $|0\rangle$ or $|1\rangle$.
- A dynamic Bloch sphere visualisation representing the current state vector.
- The spherical coordinates (θ, ϕ) corresponding to the Bloch vector.

Importantly, only the probabilities of collapsing to $|0\rangle$ and $|1\rangle$ are displayed, not the complex amplitudes, for ease of understanding. The state is instead parametrised by the (θ, ϕ) coordinates of the Bloch sphere.

Whenever the player collides with a coin, enemy, or fails to make a jump, a projective measurement in the computational (Z) basis is performed, collapsing the state probabilistically into either $|0\rangle$ or $|1\rangle$ according to the current probabilities. Players can also manually trigger a measurement by pressing the `M` key, allowing them to reset their state without needing to interact with an external object.

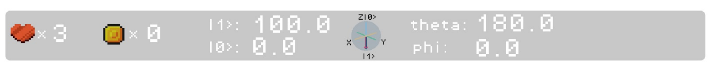


Figure 4: First dialogue by *Merlin*.

Interactions with objects in the environment, such as the on-screen guide *Merlin* for dialogue, or puzzle blocks (from Level 1 onwards), are performed by pressing the `G` key. The interactions with *Merlin* often trigger messages that help players connect their in-game actions to quantum principles. For instance, after the user has used the `X` or `Y` keys cross a section of the level, *Merlin* will appear to explain what superposition is: "While you were doing rotations, you were neither in state $|0\rangle$ nor state $|1\rangle$ but in a combination of both states with a certain probability of being in each. That is what we mean by being in superposition." Such narrative cues serve an educational role, reinforcing learning through experience.

Thus, the integration of classical platforming mechanics with quantum state dynamics allows players to engage with foundational quantum phenomena such as superposition, rotation, and measurement in an interactive and intuitive way. Gameplay thus becomes a direct exploration of quantum behaviour, requiring players to manipulate probabilities strategically to progress through levels.

4.3 Level 0: Superposition Check Zone

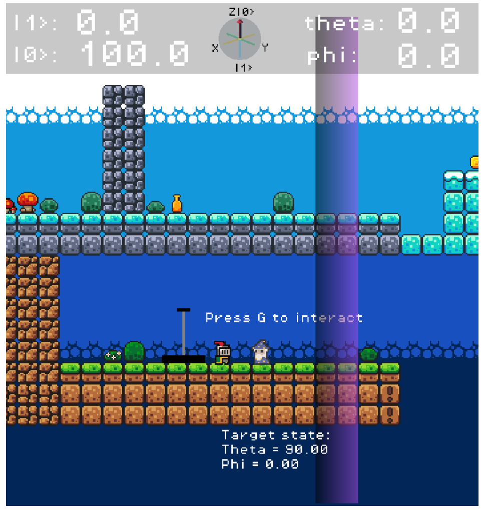


Figure 5: The *Superposition Check Zone* game mechanic.

Level 0 is intentionally designed to be mechanically simple and intuitive for the user, since this is where we introduce the core gameplay mechanics as described in the previous subsection. This level includes simple jumps and basic enemies such as slimes and shooters, which the player must avoid to progress.

In addition to these mechanics, we have also introduced the *Superposition Check Zone*. This is a special zone that requires the user to create a particular quantum state (up to a 99% fidelity) to pass through. The desired state is expressed in terms of the parameters (θ, ϕ) as seen in Fig 5, which the user is already familiar with from earlier interactions and is visible in the Heads-Up Display (HUD). When the player successfully prepares a state within the acceptable range, the zone turns green, providing clear visual feedback that passage is now permitted.

The fidelity between two pure quantum states $|\psi\rangle$ and $|\phi\rangle$ is given by [NC10]:

$$F(|\psi\rangle, |\phi\rangle) = |\langle\psi|\phi\rangle|^2$$

which is always between $[0, 1]$. This value is also displayed on the screen after the player enters and triggers the *Check Zone*.

The *Superposition Check Zone* is designed to help players develop an intuitive understanding of the Bloch sphere and explore preparing specific quantum states, in contrast to the more measurement-based mechanics featured earlier in the level.

4.4 Level 1: Circuit Puzzles and Quantum Teleportation

Level 1 builds upon the foundational mechanics introduced in Level 0, introducing two major gameplay features: *Circuit Puzzles* and *Quantum Teleportation*.

Circuit Puzzles A circuit puzzle consists of a set of wires, blocks, quantum gates, and a specified target state that the player must achieve by correctly placing gates on the blank wire slots. Both the current state and the target state are displayed below the circuit, with the current state updating in real time whenever a gate is added or removed.

Gates are scattered throughout the environment and can be collected by standing on top of them and pressing the `G` key. Upon collection, an audio cue confirms successful pickup, and the currently held gate is shown in the Head-Up Display (HUD). Similarly, placing a gate in the circuit also triggers a short audio confirmation.

To interact with the circuit puzzle, the player can place, remove, or manage quantum gates using specific blocks in the environment (as can be seen in Fig 6) by pressing `G` while standing near them:

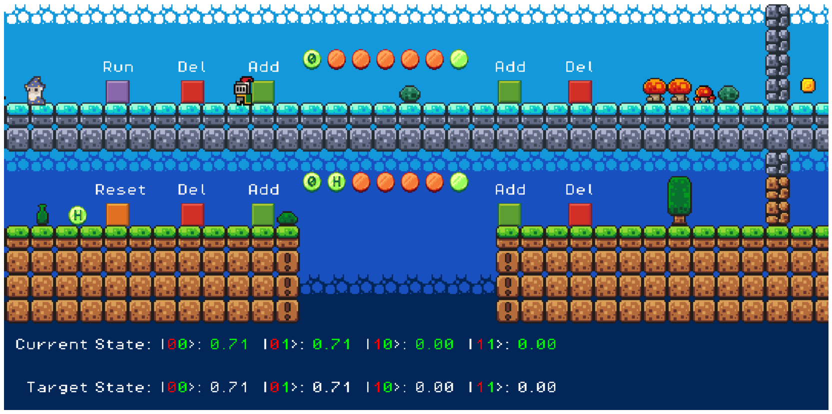


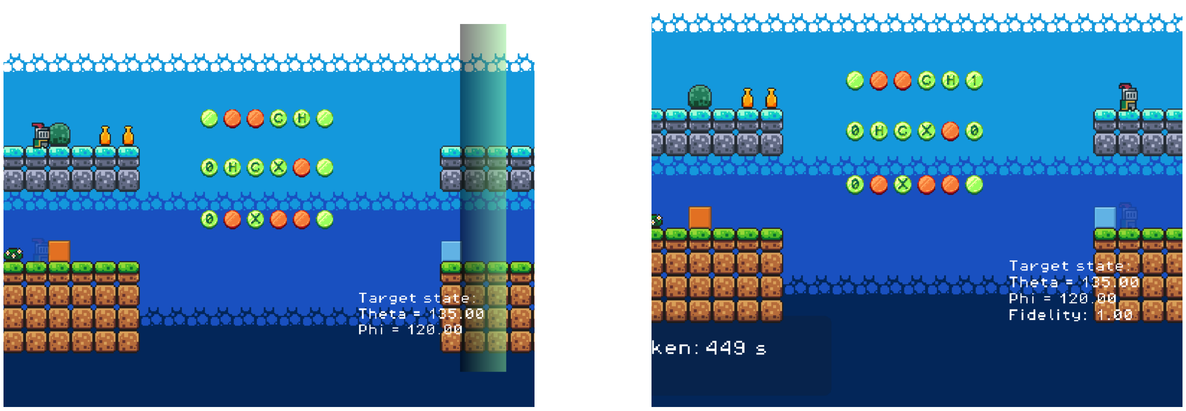
Figure 6: Puzzle view when the correct solution has been reached. The Hadamard gate is provided in the bottom left to complete the puzzle.

- **Add Block (green):** Places a collected gate on the corresponding wire.
- **Delete Block (red):** Removes a gate from the wire.
- **Reset Block (orange):** Resets the entire circuit to its initial state.
- **Run Block (purple):** Executes the circuit and checks whether the resulting state matches the target.

When the circuit executes successfully, obstacles blocking the player's path in both layers are cleared. This mechanic introduces players to fundamental quantum gates, including the X, Y, Z, H (Hadamard), and CNOT gates, through a series of puzzles.

To prevent unnecessary complexity, only the gates required for a given puzzle are made available in that area. *Merlin* appears near puzzle areas to provide brief explanations related to the gates relevant to that stage, helping the player connect gameplay actions to quantum mechanical concepts.

As the current state evolves during gate placement or removal, the amplitudes of the affected basis states are highlighted in yellow, providing a clear visual cue of which components of the quantum state are changing. When the target state is achieved successfully, all amplitudes turn green, offering immediate visual confirmation that the puzzle has been solved, as can be seen in Fig 6.



(a) Player initiating quantum teleportation.

(b) Successful teleportation to the destination.

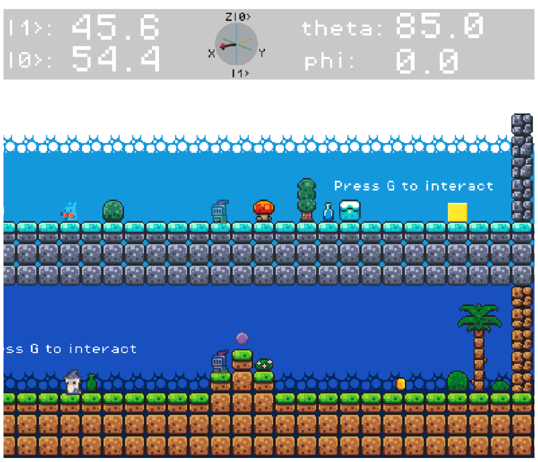
Figure 7: The Quantum Teleportation mechanic. The player prepares a superposition and successfully teleports across the gap.

Quantum Teleportation In addition to circuit puzzles, Level 1 also introduces the concept of quantum teleportation. At specific points in the level, the player encounters teleportation zones designed to simulate the transfer of quantum information across space. To cross wide gaps, the player must first prepare an appropriate superposition state and then press `G` to initiate teleportation.

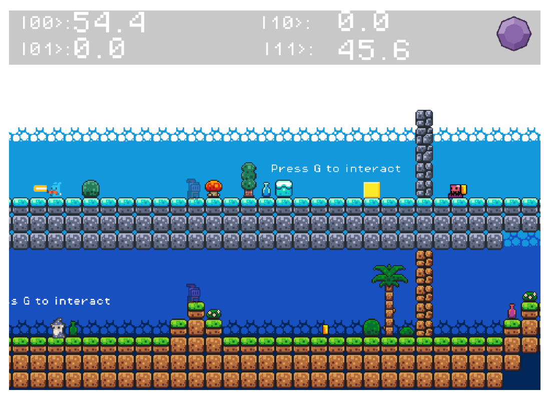
The destination area includes a *Superposition Check Zone* that validates the player's quantum state after teleportation, ensuring that the correct state has been reconstructed. In the background, the complete quantum teleportation circuit is displayed as an easter egg, including the measurement outcomes of the classical bits used in the teleportation protocol. Once the player arrives at the destination, the corresponding corrective operations (X and Z rotations) are automatically applied to their state based on these classical measurement results, faithfully mirroring the post-measurement correction step in the actual quantum teleportation process.

4.5 Level 2: Two-Qubit Systems and Entanglement

Level 2 significantly expands the gameplay by introducing two-qubit quantum systems, allowing players to interact not only with their own quantum state but also with the state of another object in the environment, such as a *Quantum Gem* or a *Qubit Enemy*. Through this level, players build upon their prior understanding of the CNOT gate from earlier circuit puzzles and also explore how single-qubit operations act within a two-qubit state space. This introduces them to the concept of entanglement and how such correlations manifest in measurement outcomes.



(a) Player in a superposition before performing the *CNOT* gate.



(b) Player in a two-qubit system after the *CNOT* gate.

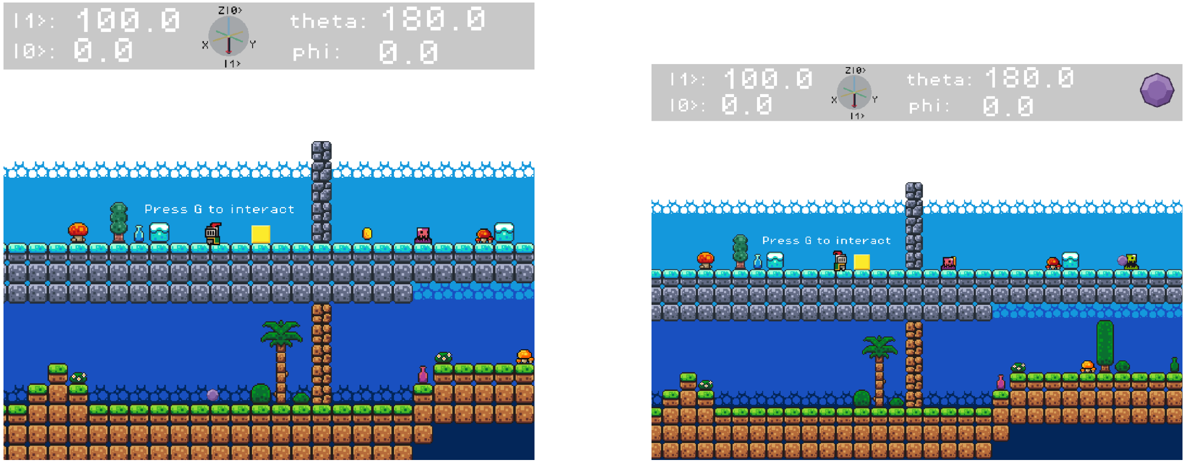
Figure 8: Creating a two-qubit system with the *Quantum Gem*.
The Heads-Up Display (HUD) changes from single-qubit to two-qubit mode.

The Quantum Gem The player first encounters a purple gem that represents a second qubit. The gem's layer position corresponds to its quantum state, the lower layer being the state $|0\rangle$ and the upper layer being the state $|1\rangle$.

The player must first prepare their own state using single-qubit rotations and then stand on the gem to perform a CNOT gate by pressing `C`. In this interaction:

- The player always acts as the control qubit, and
- The gem (or enemy) always acts as the target qubit.

Once the CNOT is performed, the player and the gem together form a *two-qubit system*, as can be seen in Fig 8. The Heads-Up Display (HUD) updates to display four probabilities corresponding to the computational basis states $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$, allowing players to visualise the evolution of their joint state. When the user presses the single-qubit rotation gates, it performs a single-qubit rotation on only the player's qubit, not on the gem.



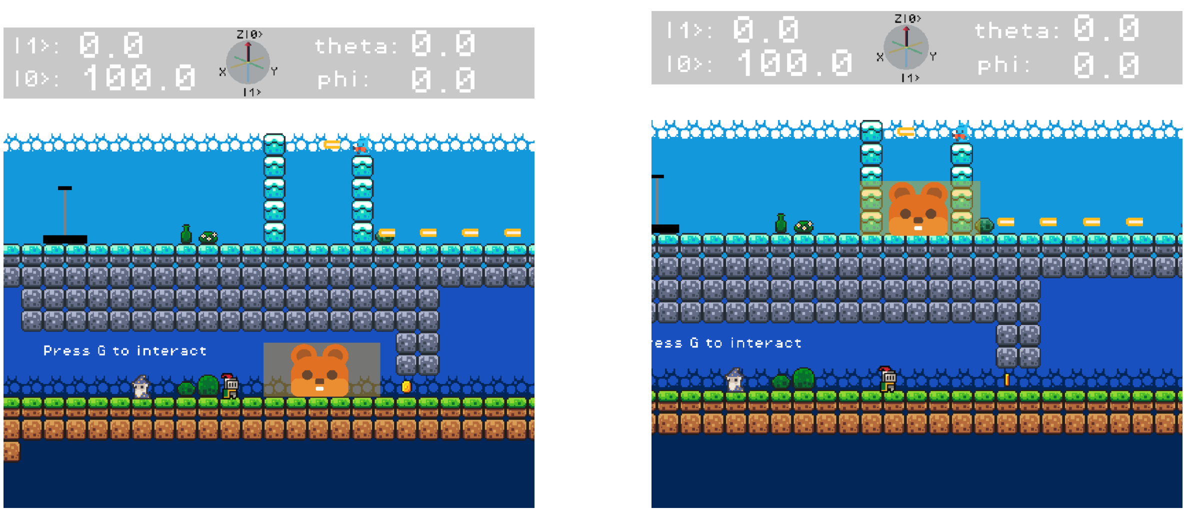
(a) A two-qubit system collapsed to state $|10\rangle$.

(b) A two-qubit system collapsed to state $|11\rangle$.

Figure 9: Measurement of a two-qubit system (the *Quantum Gem*).
This leads to the gem being either held by the player (when collapsed to $|00\rangle$ or $|11\rangle$), or the gem spawning in the opposite layer of the player (when collapsed to $|01\rangle$ or $|10\rangle$).

When a measurement occurs, both qubits collapse together into one of these four states. If the player and the gem collapse onto the same layer ($|00\rangle$ or $|11\rangle$), the gem is physically carried along with the player as they move. However, if they collapse to different layers ($|01\rangle$ or $|10\rangle$), the gem spawns in the opposite layer as the player. An example for each of these cases can be seen in Fig 9.

Attempting to re-enter superposition while carrying the gem causes it to drop, a gameplay mechanic designed to encourage exploration of measurement outcomes and state resets. Once the player holds the gem (i.e. after the two-qubit system collapses to either $|00\rangle$ or $|11\rangle$), the user needs to take the gem to a yellow block and press `G` to remove obstacles present on both layers.



(a) Qubit Enemy blocking the player's path.

(b) Path is unblocked by collapsing into $|01\rangle$.

Figure 10: The Qubit Enemy
If the two-qubit state collapses to either $|00\rangle$ or $|11\rangle$, the Qubit Enemy kills the player.

The Qubit Enemy As players advance, they encounter a new type of obstacle: the Qubit Enemy, which also represents a second qubit. The player can enter a two-qubit mode with this enemy by pressing `C` while standing in the orange area around it. Here again, the player acts as the control and the enemy as the target during a CNOT operation. An example of the Qubit Enemy can be seen in

Fig 10.

Upon measurement, the outcome determines the enemy's behaviour. If the system collapses to $|00\rangle$ or $|11\rangle$ (same layer), the enemy spawns at the player's location, killing it. If it collapses to $|01\rangle$ or $|10\rangle$ (different layers), the enemy can not kill the player.

Unlike with the gem, the user's objective here is to avoid correlated outcomes ($|00\rangle$ and $|11\rangle$) and instead aim for $|01\rangle$ or $|10\rangle$, using strategic control over their state to influence the measurement results. In some puzzles, enemies must be placed deliberately on pressure plates to open new paths.

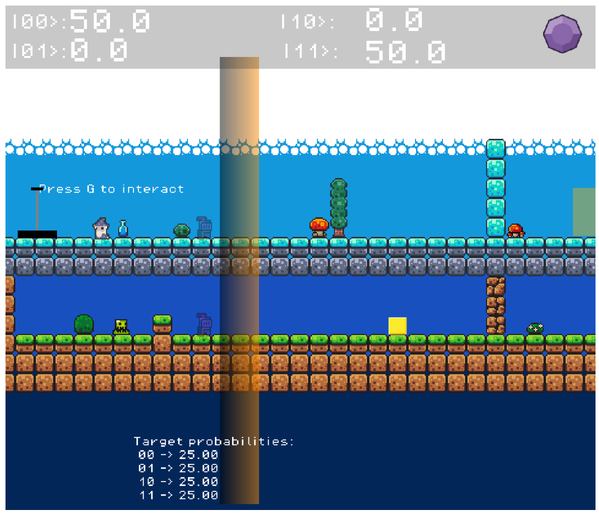


Figure 11: The *Two-Qubit Probability Check Zone* game mechanic.

Two-Qubit Probability Check Zone Later in the level, players encounter the *Two-Qubit Probability Check Zone*, similar to the single-qubit *Superposition Check Zone* introduced in Level 0. These zones require the player to prepare specific probability distributions over the four basis states to pass through. The zone turns green when the player's two-qubit state satisfies the required condition, visually confirming success. This mechanic challenges players to combine their understanding of single-qubit rotations in a two-qubit space to engineer precise quantum probability distributions.

The metric used for computing the similarity between two probability distributions is the Bhattacharyya coefficient [Bha43], which is given by:

$$B(P, Q) = \sum_i \sqrt{P_i Q_i}$$

where $P = \{P_i\}$ and $Q = \{Q_i\}$ are the two normalized probability distributions being compared. The coefficient takes values between 0 and 1, where $B = 1$ indicates identical distributions, and $B = 0$ indicates no overlap.

Understanding Entanglement The final part of the level reveals that several of the states created during gameplay, such as those with non-zero probabilities for both $|00\rangle$ and $|11\rangle$ and zero for other basis states, are in fact entangled states. Through the dialogue with *Merlin*, players learn that entanglement occurs when the two-qubit system can no longer be written as a separable product state $|x\rangle\,|y\rangle$.

The *Merlin* also explains:

- A state with equal probabilities for $|00\rangle$ and $|11\rangle$ is a *correlated entangled state*, where both qubits share the same outcome upon measurement.
- A state with probabilities for $|01\rangle$ and $|10\rangle$ represents *anti-correlated entanglement*, where the qubits always end up in opposite states.

This concept is illustrated intuitively within the game, as players observe how their actions on one qubit instantaneously affect the other, embodying the essence of quantum correlation and non-separability.

Figure 12: Entanglement is explained via dialogue.

5 Teamwork

5.1 Team Coordination and Meetings

Our team maintained a consistent meeting structure throughout the project to ensure steady progress and clear communication. We met three times every week:

- **Meeting with Prof. Christian Andersen** (Monday, 11:00–11:30 am): These sessions focused on discussing collaboration improvements to ensure a smooth workflow in the team.
- **Internal Team Meeting** (Monday, following the previous meeting): During these sessions, we reviewed progress from the previous week, discussed the tasks ahead, and assigned clear responsibilities to each member.
- **Meeting with Prof. Evert van Nieuwenburg** (Tuesday, 12:00–12:30 pm): These meetings centered on presenting the game's latest developments and receiving targeted feedback. Prof. van Nieuwenburg also provided valuable suggestions for improving the educational aspects of the game and connected us with Vincent Koeman (from Julia Cramer's group), who assisted in the design of the survey used for our analysis.

The remainder of the week was devoted to task implementation, individual development, and testing. Since we had a consistent structure, we did not face any major issues, and we finished all the required tasks by their respective deadlines.

5.2 Roles and Responsibilities

Each team member had distinct responsibilities that contributed to the project's overall development:

- **Raghav:** Developed the initial version of the platformer, introduced the superposition mechanics, and implemented circuit-based puzzles and the quantum teleportation feature. He designed Levels 0, 1, and the Challenge Level, and was also responsible for creating and conducting the player survey.
- **Mankrit:** Implemented gameplay features such as the shooter enemies, the Superposition Check Zone, the Two-Qubit Probability Check Zone, and the entanglement mechanics involving gems, enemies, and pressure plates. He also added feedback sounds, refactored major parts of the codebase and designed Level 2 of the game.
- **Aron:** Focused on user interface and gameplay polish, including adding character opacity effects, the start, pause, and loading screens, and creating the first version of Level 0. He implemented Checkpoints and the Finished game mechanics, and also helped in debugging and refining gameplay mechanics.

5.3 Lessons Learned

One of the key lessons from this project was the importance of structured planning and coordination. Early in the process, we held discussions about what could realistically be achieved within the available time frame. Although we initially considered implementing additional mechanics, our planning phase helped us recognise that focusing on the core concepts like superposition, measurement, and entanglement; would lead to a stronger and more cohesive final product.

We also initially considered conducting two rounds of surveys to iteratively improve the game based on early feedback. While this would have been beneficial in principle, our discussions made it clear that this approach was not feasible within the project's timeline, and we therefore opted to conduct a single final survey instead.

This proactive planning allowed us to stay on track from the beginning. By dividing responsibilities clearly during our weekly meetings and maintaining open communication, we ensured that development milestones were met consistently and without major setbacks. Every feature and level was completed within the planned deadlines, and the game evolved steadily in functionality.

We also encountered and resolved technical challenges along the way. For example, early on, merge conflicts occasionally led to reverting parts of existing commits due to cache files generated by the Godot engine. To address this, we protected the `main` branch and introduced a structured pull request workflow with mandatory reviews before merging. This not only prevented further versioning issues but also reinforced collaborative discipline within the team.

Overall, the project demonstrated how effective teamwork that is grounded in regular communication, realistic planning, and mutual accountability can ensure smooth progress even in a technically demanding interdisciplinary project.

6 Interdisciplinary Aspects

Our project draws upon multiple disciplines, including physics, computer science, game design and educational theory to create an interactive and scientifically correct learning experience. The major interdisciplinary components are outlined below.

- **Physics:**
  - Translating fundamental quantum computing principles into interactive, visual, and gameplay-based abstractions.
  - Identifying which physical concepts lend themselves most effectively to gameplay representation, including:
    - Superposition
    - Measurement
    - Quantum Circuits
    - Quantum Teleportation
    - Two-qubit Systems
    - Entanglement
  - Implementing the mathematical framework for single- and two-qubit systems, including state evolution via single-qubit rotations ($\mathrm{R_X}$, $\mathrm{R_Y}$, $\mathrm{R_Z}$) and entangling operations such as the CNOT gate.
  - Performing probabilistic measurements for both single- and two-qubit states, incorporating randomness consistent with quantum postulates.
  - Creating dynamic Bloch sphere visualisations that reflect the player's current qubit state in real time.
  - Computing the Fidelity and Bhattacharyya Coefficient to quantitatively evaluate how closely the player's prepared state matches the target state in the *Check Zones*.
  - Simulating the effect of adding or removing quantum gates (X, Y, Z, H, and CNOT) on the quantum circuit states.
- **Game Development (Computer Science):**
  - Implementing the game mechanics in a scalable, modular, and optimally designed manner.

  - Structuring the codebase to promote reusability and flexibility, ensuring future levels and features can be integrated efficiently.
  - Managing player state, physics, collisions, and in-game logic using Godot's framework and GDScript.
  - Ensuring numerical stability and computational efficiency when simulating quantum systems in real time.

- **Game Design:**
  - Designing levels that are intuitive, engaging, and conceptually reflective of quantum principles rather than purely classical challenges.
  - Incorporating platforming mechanics such as *coyote time* (a short grace period allowing the player to jump after leaving a platform), ensuring smoother gameplay.
  - Balancing conceptual and mechanical difficulty, ensuring the player can grasp challenging quantum concepts without being hindered by overly complex controls.
  - Strategically placing coins, enemies, and checkpoints to fine-tune each level's learning curve and difficulty progression.

- **UI/UX Design:**
  - Communicating abstract quantum states and transitions visually, for instance using opacity to represent measurement probabilities and adding a dynamic Bloch sphere visualisation.
  - Providing multisensory feedback like sound effects for coin collection, successful measurements, and entering two-qubit modes; to enhance immersion and intuitive understanding.
  - Providing in-game feedback like *Check Zones* turning green when the player is in an acceptable state and *Circuit Puzzles* where the amplitudes turn green when arriving at the correct solution.
  - Maintaining a consistent and minimalistic Heads-Up Display (HUD) that presents essential parameters such as probabilities, (θ, ϕ) values, and Bloch sphere visualisation without visual clutter.

- **Education:**
  - Ensuring that the introduction of quantum concepts is both scientifically accurate and pedagogically effective, enabling players to build intuition through active experimentation.
  - Reinforcing conceptual understanding through in-game dialogues and feedback. For example, players are explicitly informed when they have successfully created a superposition or entangled state, connecting gameplay to underlying physics.
  - Using contextual guidance (e.g., dialogue from Merlin) to demystify abstract concepts like measurement, entanglement, and quantum gates, ensuring that educational goals are seamlessly integrated with gameplay progression.

7 Survey Analysis

In this section, we will first discuss the methodology of the survey, and then we will analyse its results and discuss some of its nuances.

7.1 Survey Design and Methodology

As we have mentioned in section 2.2, we conducted a survey to get scores against our success criteria. While certain aspects of the game can be measured objectively, others, such as enjoyment of the game and how well someone understood the theory, are subjective, which is why we decided to create the survey.

When creating the survey, we wanted to avoid biases in survey design[Sur], which is why we tried to avoid a lot of neutral options, described what each option meant, avoided leading and loaded questions, avoided using jargon as much as possible, and used balanced rating scales. We then discussed this with Vincent to obtain practical feedback and insights from his experience, as mentioned earlier in Section 5. We used the IMPACTLAB question bank [IMP] as inspiration for some questions.

In our survey, we first ask questions related to the general background of the respondent, like prior knowledge, how they found the game, the highest level tried, etc. The aim of these questions is to:

1. Make groups based on how much they know about quantum computing beforehand.
2. Only show them questions up to the highest level they tried.
3. Notice any possible biases to keep in mind.

We then ask questions regarding our success criteria. These are asked for the game in general, and for each level as well. Some examples include:

- **Gameplay Engagement:** How did this game make you feel?
- **Conceptual Intuitiveness:** How well do you think you understand the superposition and measurement game mechanics?
- **Scientific Accuracy:** How accurately was superposition represented?
- **Technical Implementation:** How would you rate the overall visual style and atmosphere of the game?

Questions related to Gameplay Engagement and Technical Implementation are asked to all participants. For Scientific Accuracy, we only consider the results of those who knew quantum before playing the game. For Conceptual Intuitiveness, we only consider the results of those who did not know quantum before playing the game. To account for this, we would conditionally show certain questions depending on the highest level tried and how much they knew about quantum before playing the game. We decided to make the survey using Qualtrics since it was free using our Leiden University account and it handled branching logic very conveniently.

Most of our questions are 5-point Likert scale questions, i.e., questions with five options. We convert these to numbers from 1-5 where 1 is the most negative and 5 is the most positive response. For Conceptual Intuitiveness, we also ask people who didn't know quantum before playing the game to describe certain phenomena like superposition, etc. These are also then graded on the same scale of 1-5, with 1 being completely inaccurate and 5 being extremely accurate. For each criterion, we ask questions about each level specifically, and for some criteria, we also ask questions relating to the overall game. Although the overall game questions differ from the level-specific ones, they still address the same underlying evaluation criterion. In the end, for each success criterion, we average the results for each question, weighted by the number of responses the question got.

The survey was open from 15 October 2026 to 27 October 2026. In that time, the game got 667 views, 289 browser plays and the survey got 56 responses.

7.2 General Background Questions

In this section, we will discuss the questions in the survey to get the general background of the respondents.

The first thing we needed to know was how much the user knew about quantum computing before playing the game. As can be seen in figure 13, the highest number belongs to "Heard of it from news/friend/etc" with 19 responses, and the least is "Never heard of it" with a single response. We have also broadly grouped the first 3 options as "Don't know quantum computing" and the last 3 options as "Know quantum computing". From figure 14 we can see that the 2 groups have a roughly similar number of responses, which means it will be less likely for one group to dominate the results or skew the overall trends.

We then asked the participants their age and where they were from, but these were not compulsory questions to respect their privacy. Most responses belong to 20-30 years old, with few responses in the <20 and >50 years old brackets and none in the range 30-50. The majority of our audience were in

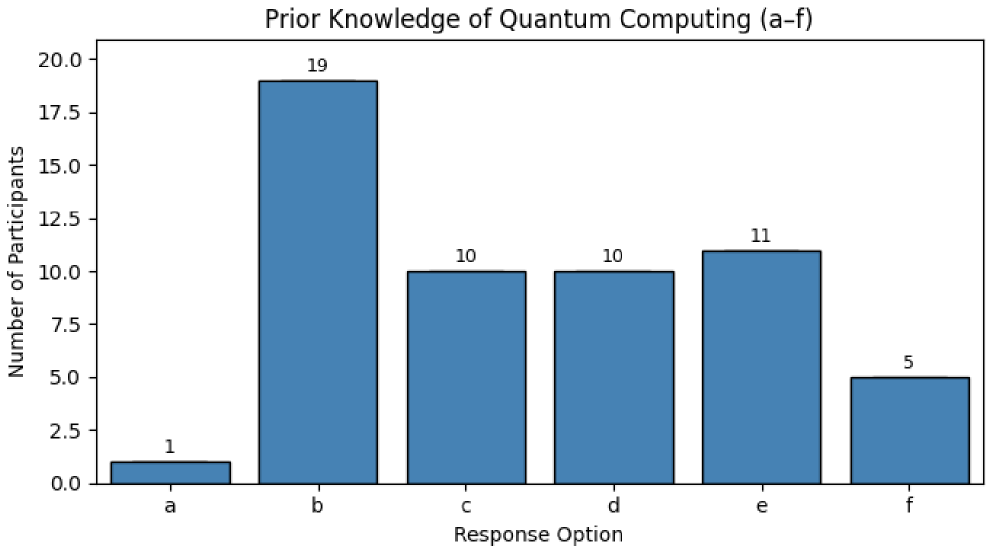


Figure 13: Prior knowledge of quantum computing. The x-axis labels correspond to: (a) Never heard of it, (b) Heard of it from news/friend/etc, (c) Have watched YouTube video/read basics online, (d) Have taken a course on Quantum Mechanics/Quantum Computing, (e) Have/Currently following a program related to quantum topics, and (f) Doing research in quantum related fields.

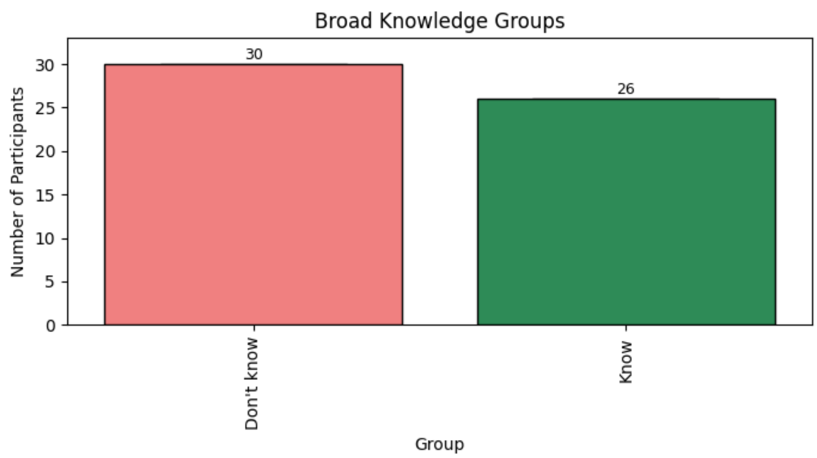


Figure 14: Broader categories of prior quantum knowledge into "Don't know" and "Know" quantum computing.

the age range of 20-30, which does have some overlap with the target audience that we expect to be able to understand and learn from the game. We also noticed that most people discovered the game via recommendation from friends/colleagues/etc.

Most people were from India, with the second most from the Netherlands. Since our group consisted of two people who originated from India and one from the Netherlands, this fact, combined with how people discovered the game, leads us to conclude that most of the respondents are known either directly or indirectly by us, which may lead to some bias in their responses. While we cannot really account for that in calculating our scores, we do need to keep in mind that our results could be skewed towards the positive side.

As expected from figure 15, most people only tried level 0 and stopped, with the number decreasing for higher levels, but fortunately, the number of responses in either knowledge level is almost the same for each further level. In total, 56 people have played level 0, 33 played level 1, 14 played level 2, and 3 played the challenge level. Since the challenge level does not introduce any new mechanics and very few people have tried it, we will not be including its results when calculating our success criteria.

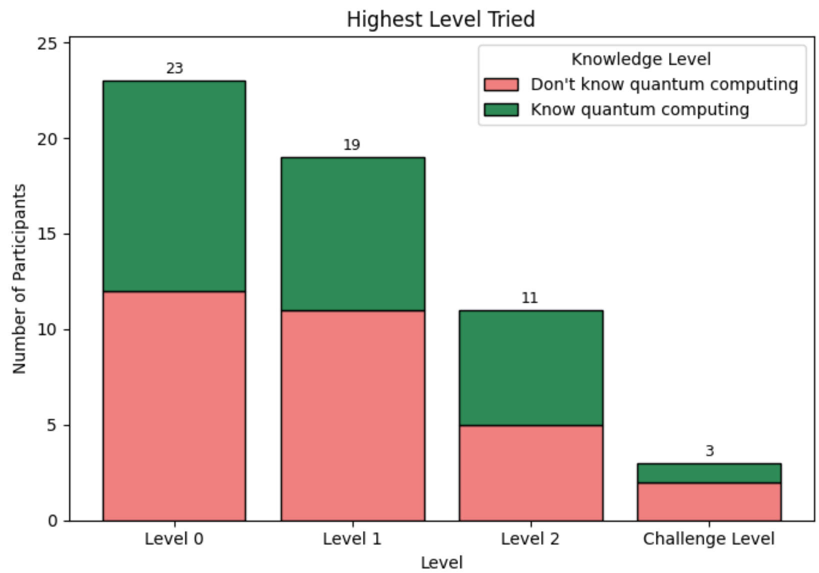


Figure 15: Distribution of participants based on the highest level they tried. The x-axis represents the highest level attempted by each participant.

7.3 Gameplay Engagement

In this section, we will be discussing the questions related to gameplay engagement and how much the respondents enjoyed the game. We will also be looking at the difficulty they felt while playing the game. We will first look into the responses for the entire game in general, then for each level, and finally, discuss the responses.

7.3.1 Game In General

The responses can be seen in figure 16. As mentioned in section 7.1, to get a score, we gave the options a value of 1 to 5 from left to right. Thus, for difficulty, we would like a score around 3, but for the other questions on gameplay engagement, the higher the score, the more positive our result.

Evaluation Topic	**Don't Know Quantum**	**Know Quantum**	**Overall**
How did this game make you feel?	3.77	4.15	3.95
How likely are you to recommend this game or play similar games in the future?	3.70	3.81	3.75
Would you like to keep playing this game or see more levels added in the future?	3.33	3.33	3.33
Gameplay Engagement Score (avg. of above 3)	**3.60**	**3.76**	**3.68**
How difficult was the game?	3.27	2.83	3.07

Table 1: Average values for responses to questions related to Gameplay Engagement for the overall game. The "Gameplay Engagement Score" represents the average of the first three questions.

From table 1, we get an overall average of 3.07 for the difficulty of the game in general. To get a score for gameplay engagement for the game in general, we average the values for the other three questions, giving us an overall average of 3.68.

7.3.2 Level 0

The responses can be seen in figure 17. From table 2, we get an overall average of 2.48 for the difficulty of level 0 and a gameplay engagement score of 3.61.

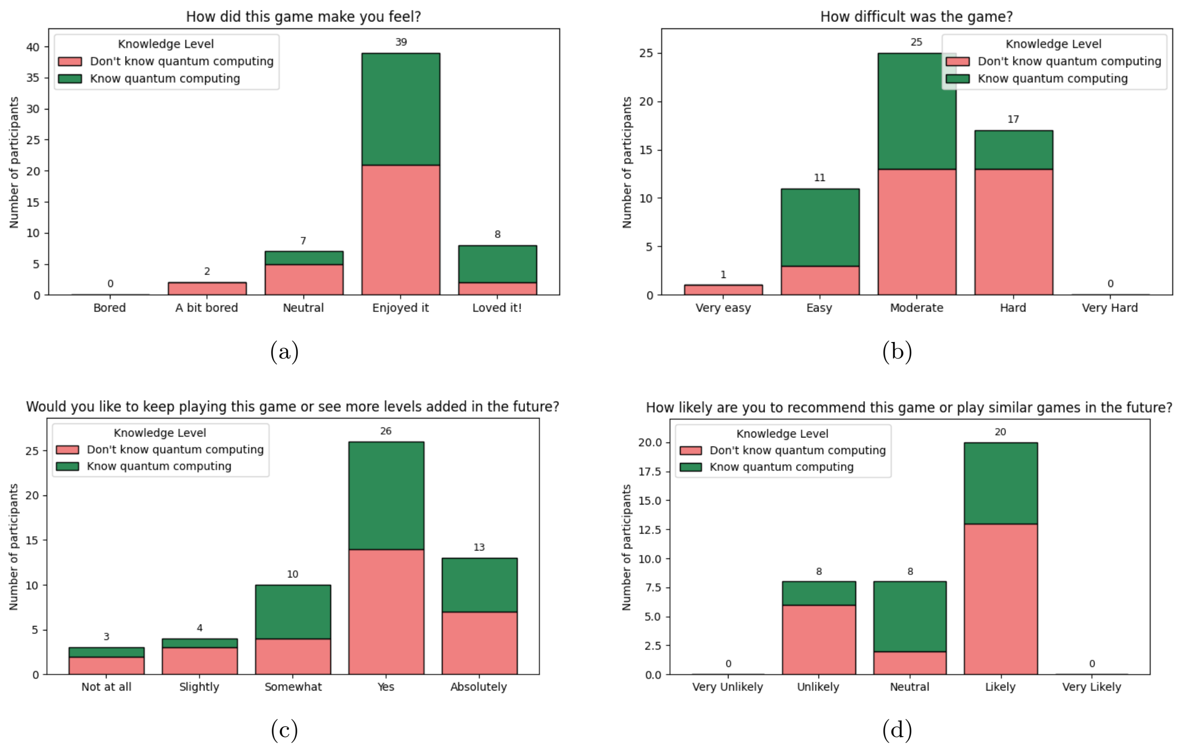


Figure 16: Comparison of player responses across gameplay engagement questions for the overall game: (a) How did this game make you feel? (b) How difficult was the game? (c) Would you like to keep playing this game or see more levels added in the future? (d) How likely are you to recommend this game or play similar games in the future?

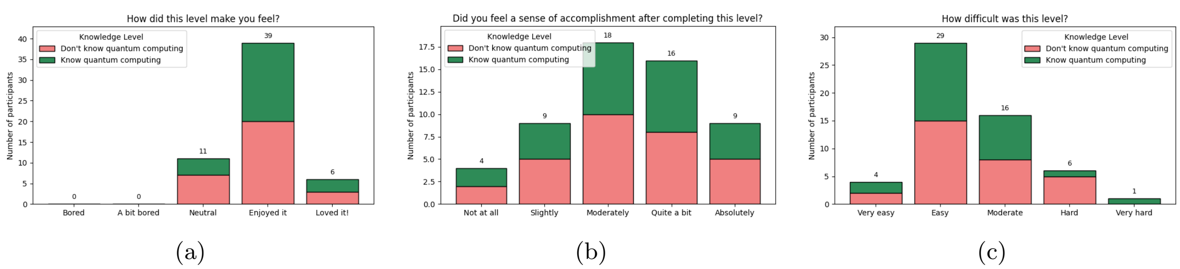
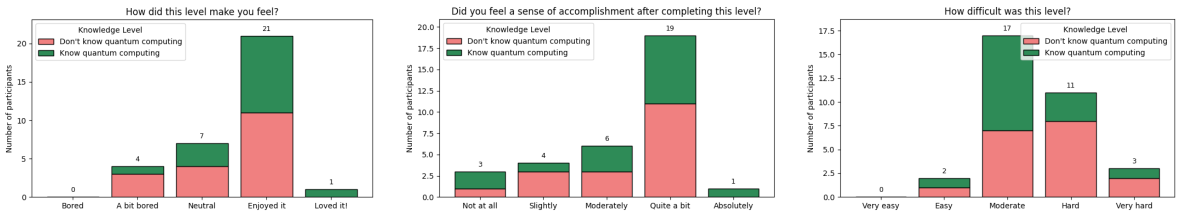


Figure 17: Comparison of player responses across gameplay engagement questions for level 0: (a) How did this level make you feel?, (b) Did you feel a sense of accomplishment after completing this level?, and (c) How difficult was this level?

Evaluation Topic	Don't Know Quantum	Know Quantum	Overall
How did this level make you feel?	3.87	3.96	3.91
Did you feel a sense of accomplishment after completing this level?	3.30	3.31	3.30
Gameplay Engagement Score (avg. of above 2)	**3.58**	**3.64**	**3.61**
How difficult was this level?	2.53	2.42	2.48

Table 2: Average values for responses to questions related to Gameplay Engagement for level 0. The "Gameplay Engagement Score" represents the average of the first three questions.

7.3.3 Level 1



(a) (b) (c)

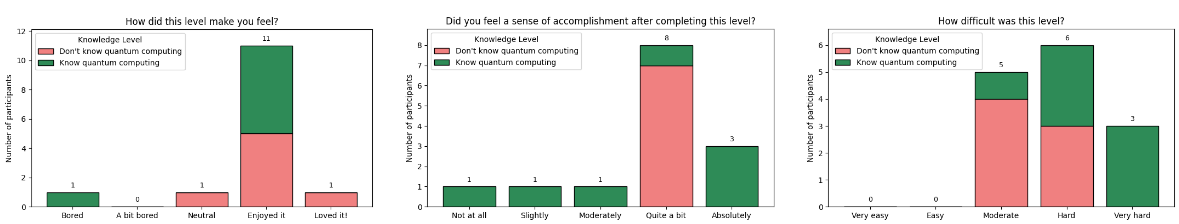
Figure 18: Comparison of player responses across gameplay engagement questions for level 1: (a) How did this level make you feel?, (b) Did you feel a sense of accomplishment after completing this level?, and (c) How difficult was this level?

Evaluation Topic	Don't Know Quantum	Know Quantum	Overall
How did this level make you feel?	3.44	3.73	3.58
Did you feel a sense of accomplishment after completing this level?	3.33	3.33	3.33
Gameplay Engagement Score (avg. of above 2)	**3.39**	**3.53**	**3.46**
How difficult was this level?	3.61	3.27	3.45

Table 3: Average values for responses to questions related to Gameplay Engagement for level 1. The "Gameplay Engagement Score" represents the average of the first three questions.

The responses can be seen in figure 18. From table 3, we get an overall average of 3.45 for the difficulty of level 1 and a gameplay engagement score of 3.46.

7.3.4 Level 2



(a) (b) (c)

Figure 19: Comparison of player responses across gameplay engagement questions for level 2: (a) How did this level make you feel?, (b) Did you feel a sense of accomplishment after completing this level?, and (c) How difficult was this level?

The responses can be seen in figure 19. From table 4, we get an overall average of 2.48 for the difficulty of level 2 and a gameplay engagement score of 3.61.

7.3.5 Discussion

In general, people seem to enjoy the game as for each level our average gameplay engagement score is above 3. We do notice that level 1 has a relatively lower score. In this level, there were circuit puzzles, and according to the feedback we received, the level felt tedious at times. We also note that level 2 has the highest gameplay engagement score, but also has the least number of respondents. In all cases except for level 2, the group that knew quantum before playing the game liked it slightly more than the group that did not know. The difficulty generally lies around moderate and average

Evaluation Topic	Don't Know Quantum	Know Quantum	Overall
How did this level make you feel?	4.00	3.57	3.79
Did you feel a sense of accomplishment after completing this level?	4.00	3.57	3.79
Gameplay Engagement Score (avg. of above 2)	**4.00**	**3.57**	**3.79**
How difficult was this level?	3.43	4.29	3.86

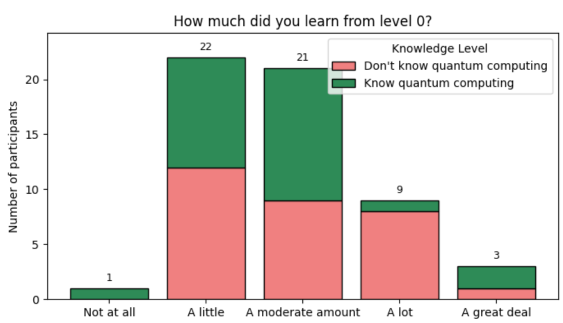
Table 4: Average values for responses to questions related to Gameplay Engagement for level 2. The "Gameplay Engagement Score" represents the average of the first three questions.

difficulty (weighted by number of responses) is 3.01, which is good since that means that players did not find it too easy nor too challenging. By computing the weighted average of the individual Gameplay Engagement Scores, the final **Gameplay Engagement Score is 3.62**, which is between **Neutral** and **Enjoyed it**.

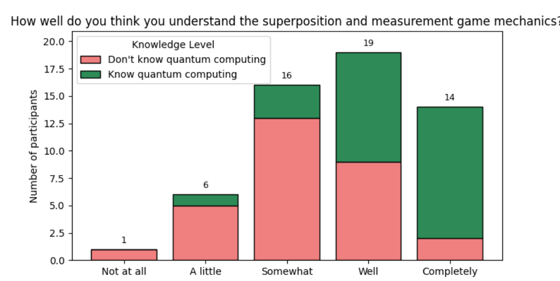
7.4 Conceptual Intuitiveness

In this section, we will be discussing the questions related to conceptual intuitiveness and how much they really understood. We will first look into the responses for each level, and then discuss the responses. While we do have responses from both the group that knows quantum and the group that does not, when computing the score, we will only be considering the responses of the group that does not know quantum computing.

7.4.1 Level 0



(a)



(b)

Figure 20: Participants' responses for (a) perceived learning from Level 0 and (b) understanding of the superposition and measurement mechanics.

We can see the responses to the Likert questions in figure 20. The text-based questions for this level are about Superposition and Measurement. One common misconception we noticed is that people believe that the higher probability is always chosen when we measure. Furthermore, a few people confused superposition and measurement with other definitions, such as measuring a quantity.

From table 5, we get an overall average of 3.04 for the conceptual intuitiveness score of level 0.

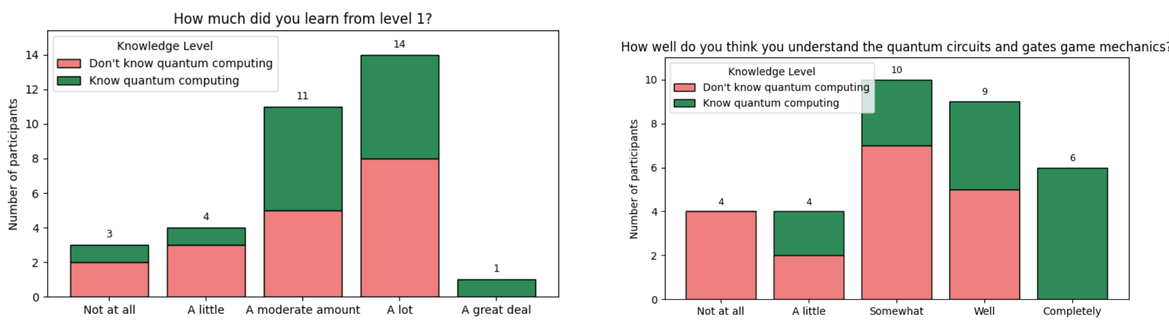
7.4.2 Level 1

We can see the responses to the Likert questions in figure 21. The text-based questions for this level are about X, Y, Z, H and CNOT gates. One common misconception we noticed is that people confused the X, Y, and Z gates for the circuit with the superposition mechanic (R_X, R_Y, R_Z). Also, quite a lot of people only talked about some of the gates.

From table 6, we get an overall average of 2.74 for the conceptual intuitiveness score of level 1.

Evaluation Topic	Don't Know Quantum (Avg.)
How much did you learn from Level 0?	2.93
How well do you think you understand the superposition and measurement game mechanics?	3.20
Avg. Likert Responses Score	**3.06**
Superposition Description Score	3.53
Measurement Description Score	2.5
Avg. Text Responses Score	**3.02**
Conceptual Intuitiveness Score	**3.04**

Table 5: Average responses for Level 0 learning and understanding questions among participants without prior quantum knowledge.



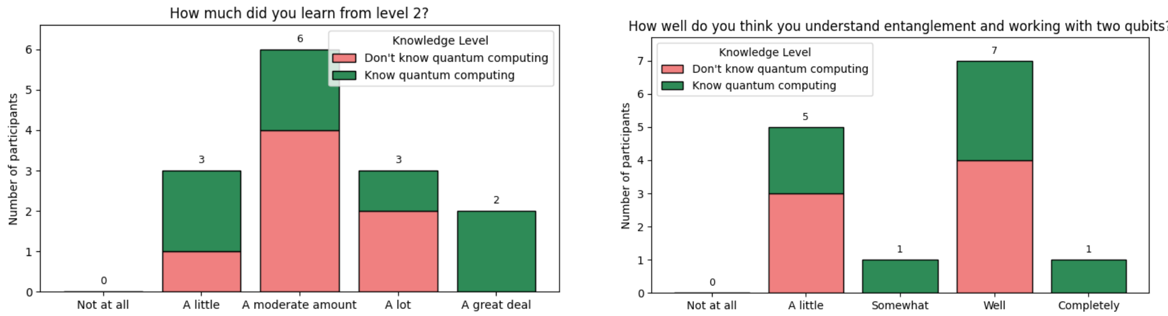
(a) (b)

Figure 21: Participants' responses for (a) perceived learning from Level 1 and (b) understanding of the quantum circuit mechanics.

Evaluation Topic	Don't Know Quantum (Avg.)
How much did you learn from Level 1?	3.06
How well do you think you understand the quantum circuits and gates game mechanics?	2.72
Avg. Likert Responses Score	**2.89**
X, Y, and Z gate Description Score	2.67
H and CNOT gate Description Score	2.53
Avg. Text Responses Score	**2.6**
Conceptual Intuitiveness Score	**2.74**

Table 6: Average responses for Level 1 learning and understanding questions among participants without prior quantum knowledge.

7.4.3 Level 2



(a) (b)

Figure 22: Participants' responses for (a) perceived learning from Level 2 and (b) understanding of the 2 qubit system game mechanics.

We can see the responses to the Likert questions in figure 22. The text-based question for this level is about Entanglement. There was no real misconception, but it is worth noting that there are only 7 entries here.

Evaluation Topic	**Don't Know Quantum (Avg.)**
How much did you learn from Level 2?	3.14
How well do you think you understand entanglement and working with two qubits?	3.14
Avg. Likert Responses Score	**3.14**
Entanglement Description Score	3.86
Avg. Text Responses Score	**3.86**
Conceptual Intuitiveness Score	**3.5**

Table 7: Average responses for Level 2 learning and understanding questions among participants without prior quantum knowledge.

From table 7, we get an overall average of 3.5 for the conceptual intuitiveness score of level 2.

7.4.4 Discussion

In general, people seem to have learned a bit from the game. Level 1 has a lower score mostly due to confusing the question about X, Y and Z gates with the rotations used to create superposition. Also, many people seem to have a major confusion with measurement, especially because it is a commonly used word. The best text response score was for the description of entanglement. Of course, keep in mind that fewer people played level 2. By computing the weighted average of the individual Conceptual Intuitiveness Scores, the final **Conceptual Intuitiveness Score is 3.00**, which means that people feel like they have learned a **Moderate** amount from the game.

7.5 Scientific Accuracy

In this section, we will be discussing the questions related to scientific accuracy. We will first look into the responses for each level and then discuss the responses. Here, we only consider the responses of the group that knows quantum computing. For the graphs in this section, the x-axis labels are the options provided to the users. They are:

a) Completely inaccurate

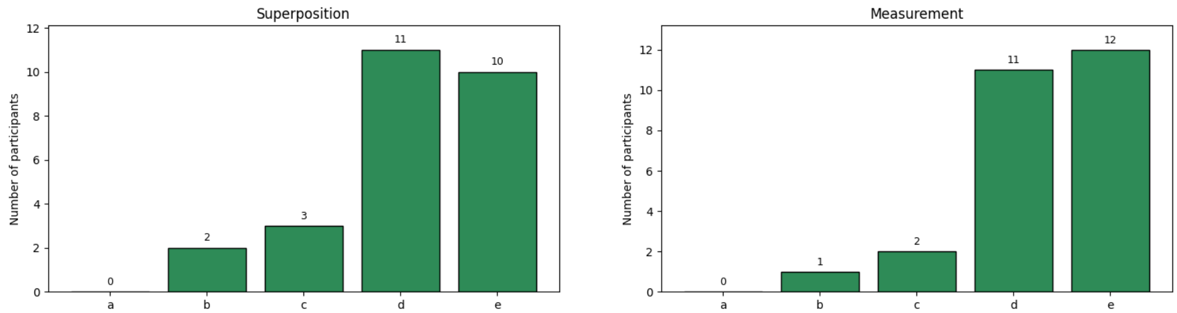
b) Misses some key aspects

c) Room for misinterpretations

d) Misses minor aspects

e) Extremely accurate

7.5.1 Level 0



(a) (b)

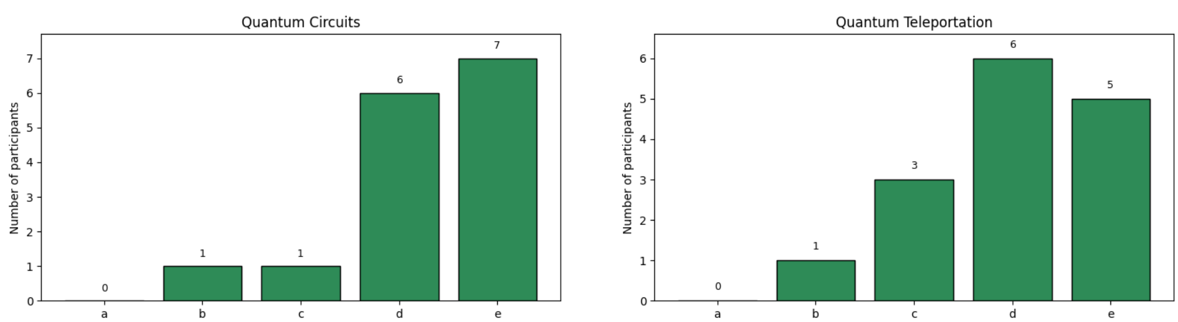
Figure 23: Participants' responses for (a) accuracy of superposition and (b) accuracy of measurement.

Evaluation Topic	**Know Quantum**
Superposition	4.12
Measurement	4.31
Scientific Accuracy Score (avg.)	**4.22**

Table 8: Average ratings for scientific accuracy questions in Level 0 among participants who reported prior knowledge of quantum computing.

For level 0, we ask about superposition and measurement. The responses can be seen in figure 23. From table 8, we get an overall average of 4.22 for the scientific accuracy score of level 0.

7.5.2 Level 1



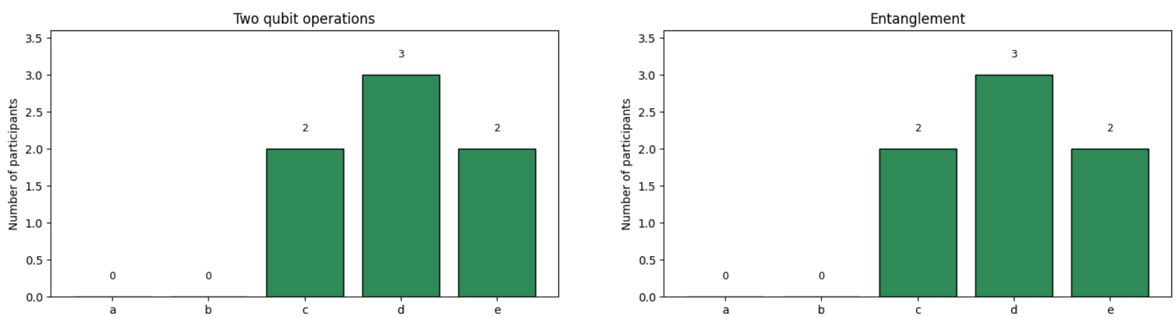
(a) (b)

Figure 24: Participants' responses for (a) accuracy of quantum circuits and (b) accuracy of teleportation.

For level 1, we ask about the quantum circuit puzzle and quantum teleportation. The responses can be seen in figure 24. From table 9, we get an overall average of 4.14 for the scientific accuracy score of level 1.

Evaluation Topic	Know Quantum
Quantum Circuits	4.27
Quantum Teleportation	4.00
Scientific Accuracy Score (avg.)	**4.14**

Table 9: Average ratings for scientific accuracy questions in Level 1 among participants who reported prior knowledge of quantum computing.



(a) (b)

Figure 25: Participants' responses for (a) accuracy of two qubit operations and (b) accuracy of entanglement.

Evaluation Topic	Know Quantum
Two Qubit Operations	4.00
Entanglement	4.00
Scientific Accuracy Score (avg.)	**4.00**

Table 10: Average ratings for scientific accuracy questions in Level 2 among participants who reported prior knowledge of quantum computing.

7.5.3 Level 2

For level 2, we ask about two qubit operations and entanglement. The responses can be seen in figure 25. From table 10, we get an overall average of 4.00 for the scientific accuracy score of level 2.

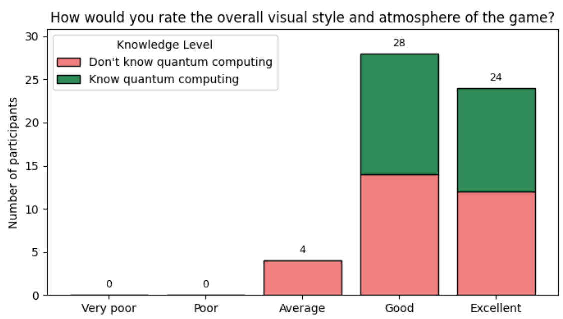
7.5.4 Discussion

In general, people believe that our game accurately represents superposition, measurement, quantum circuits, teleportation, two qubit operations and entanglement. By computing the weighted average of the individual Scientific Accuracy Scores, the final **Scientific Accuracy Score is 4.16**, which means that people feel it is accurate but **Misses minor aspects**.

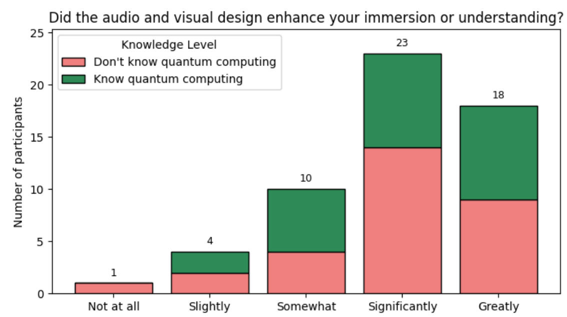
7.6 Technical Implementation

In this section, we will be discussing the questions related to technical implementation. We will first look into the responses for the entire game in general, then for each level, and then discuss the responses.

7.6.1 Game In General



(a)



(b)

Figure 26: Participants' responses for (a) How would you rate the overall visual style and atmosphere of the game? and (b) Did the audio and visual design enhance your immersion or understanding?

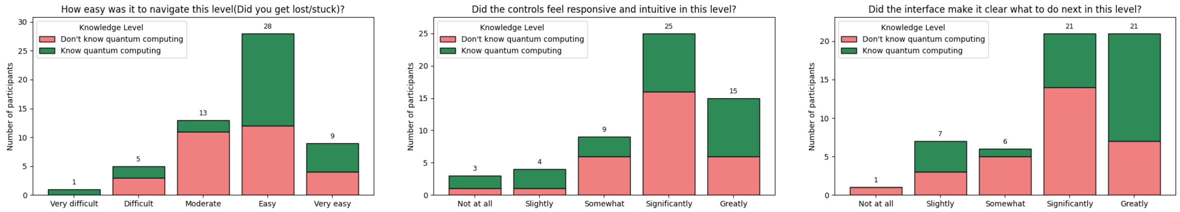
Evaluation Topic	Don't Know Quantum	Know Quantum	Overall
How would you rate the overall visual style and atmosphere of the game?	4.27	4.46	4.36
Did the audio and visual design enhance your immersion or understanding?	3.93	3.96	3.95
Technical Implementation Score (avg. of above 2)	**4.10**	**4.21**	**4.16**

Table 11: Average ratings for audiovisual and technical design aspects of the game. The "Technical Implementation Score" represents the average of the above questions.

The responses can be seen in figure 26. From table 11, we get an overall average of 4.16 for the technical implementation score of the game in general.

7.6.2 Level 0

The responses can be seen in figure 27. From table 12, we get an overall average of 3.82 for the technical implementation score of level 0.

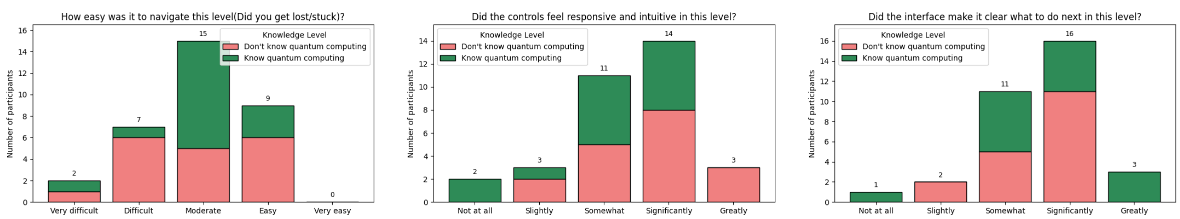


(a) (b) (c)

Figure 27: Participants' responses for (a) ease of navigation, (b) control responsiveness and intuitiveness, and (c) interface clarity in Level 0.

Evaluation Topic	**Don't Know Quantum**	**Know Quantum**	**Overall**
How easy was it to navigate this level (Did you get lost/stuck)?	3.57	3.85	3.70
Did the controls feel responsive and intuitive in this level?	3.83	3.77	3.80
Did the interface make it clear what to do next in this level?	3.77	4.19	3.96
Technical Implementation Score (avg. of above 3)	**3.72**	**3.94**	**3.82**

Table 12: Participants' responses for evaluation topics related to Level 0 interface and control quality. The "Technical Implementation Score" represents the average of the above questions.



(a) (b) (c)

Figure 28: Participants' responses for (a) ease of navigation, (b) control responsiveness and intuitiveness, and (c) interface clarity in Level 1.

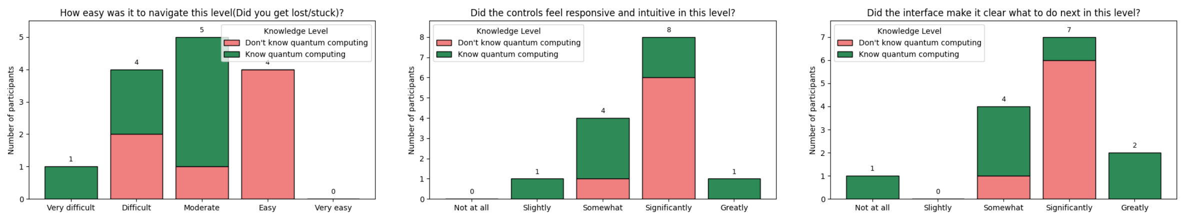
Evaluation Topic	**Don't Know Quantum**	**Know Quantum**	**Overall**
How easy was it to navigate this level (Did you get lost/stuck)?	2.89	3.00	2.94
Did the controls feel responsive and intuitive in this level?	3.67	3.07	3.39
Did the interface make it clear what to do next in this level?	3.50	3.60	3.55
Technical Implementation Score (avg. of above 3)	**3.35**	**3.22**	**3.29**

Table 13: Participants' responses for evaluation topics related to Level 1 interface and control quality. The "Technical Implementation Score" represents the average of the above questions.

7.6.3 Level 1

The responses can be seen in figure 28. From table 13, we get an overall average of 3.29 for the technical implementation score of level 1.

7.6.4 Level 2



(a) (b) (c)

Figure 29: Participants' responses for (a) ease of navigation, (b) control responsiveness and intuitiveness, and (c) interface clarity in Level 2.

Evaluation Topic	Don't Know Quantum	Know Quantum	Overall
How easy was it to navigate this level (Did you get lost/stuck)?	3.29	2.43	2.86
Did the controls feel responsive and intuitive in this level?	3.86	3.43	3.64
Did the interface make it clear what to do next in this level?	3.86	3.43	3.64
Technical Implementation Score (avg. of above 3)	**3.67**	**3.10**	**3.38**

Table 14: Participants' responses for evaluation topics related to Level 0 interface and control quality. The "Technical Implementation Score" represents the average of the above questions.

The responses can be seen in figure 29. From table 14, we get an overall average of 3.38 for the technical implementation score of level 2.

7.6.5 Discussion

In general, people are somewhat positive on the game quality. In particular, they like the graphics and audio. By computing the weighted average of the individual Technical Implementation Scores, the final **Technical Implementation Score is 3.79**, which means that people feel it is closer to **Good** technical implementation.

8 Assessment and Possible Improvements

Evaluation Topic	Don't Know Quantum	Know Quantum	Overall
Final Gameplay Engagement Score	**3.58**	**3.65**	**3.62**
Final Conceptual Intuitiveness Score	**3.00**	**-**	**3.00**
Final Scientific Accuracy Score	**-**	**4.16**	**4.16**
Final Technical Implementation Score	**3.77**	**3.81**	**3.79**
Final Score	**3.45**	**3.87**	**3.64**

Table 15: Averages of scores for all of the success criteria. The "Final Score" represents the average of all other scores.

In general, people seemed to have enjoyed the game. Some common issues we noticed include very lengthy dialogues, misunderstanding certain concepts like measurement, and some puzzles being too tedious. The compiled results can be found in table 15 with our **final score being 3.64**, which is greater than the average, i.e, 3. That being said, there is room for improvement. Some possible improvements include

- Split up the levels to slowly introduce more concepts and give the players time to understand.
- Try to reduce dialogue, spread it out a bit more and try to have more interactive explanations to aid with understanding.
- Try to focus more on distinguishing terms like measurement from the conventional interpretation.
- Improving the user interface so that users can keep up with the amount of information being given.

9 Conclusion

Quantum computing is a rapidly growing field that is often difficult to understand, as its principles are highly abstract and non-intuitive, especially to those unfamiliar with the mathematics behind it. To help explain some concepts of quantum computing, we have made a game with the aim to be fun and intuitive for the audience, while still accurately representing the phenomena. In this game, we represented superposition, measurement, rotations on a Bloch sphere, quantum circuits, quantum teleportation, two-qubit operations and entanglement.

To evaluate the effectiveness of our game, we first decided on and defined success criteria to be assessed. We then conducted a survey to get feedback on our success criteria, i.e., Gameplay Engagement, Conceptual Intuitiveness, Scientific Accuracy, and Technical Implementation.

The responses were then analysed to identify underlying trends, possible biases, patterns in participant understanding, and variations in perception across different knowledge groups. This analysis provided insight into how engaging the game was, how accurately phenomena were represented and how effectively the game conveyed quantum concepts. It also highlighted areas for potential improvement in both design and educational impact.

Overall, we received a final average score of 3.64 out of 5, which is greater than the average neutral midpoint of 3. This indicates that participants generally found the game enjoyable and informative, while maintaining scientific accuracy. Specifically

- **Gameplay Engagement:** We got an overall score of **3.62** which is between feeling **Neutral** about the game and **Enjoying it**.
- **Conceptual Intuitiveness:** We got an overall score of **3.0** which means that the users have learnt a **Moderate** amount from the game.
- **Scientific Accuracy:** We got an overall score of **4.16**, which means that the users feel the representations of phenomena only **Misses minor aspects**.
- **Technical Implementation:** We got an overall score of **3.79** which is in between **Average** and **Good** technical implementation.

Future work includes:

- Introducing more concepts like decoherence, phase, etc.
- Working on more quantum games like Quantum Angry Birds, etc.
- Building plugins so that other game developers can make games using quantum phenomena without having to code it themselves.

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10 Appendix

10.1 AI Usage

For this project, AI was used to help with code and check the grammar of the report.